Keshav Singh

+353 872292557 | keshav.singh2698@gmail.com | LinkedIn Profile | GitHub Profile | Portfolio | Ireland

PROFILE

DevOps Engineer with 4+ years of professional experience spanning cloud infrastructure, CI/CD automation, Linux systems and Java application development. Hands-on experience with AWS, Terraform, Jenkins, GitHub Actions, Docker and Kubernetes, including infrastructure provisioning, deployment automation, monitoring and production troubleshooting. AWS Certified DevOps Engineer – Professional with an MSc in Cloud Computing and independent projects in AWS cost optimization and Kubernetes diagnostics.

CORE SKILLS

Cloud & Infrastructure: AWS (EC2, S3, IAM, VPC, RDS, EKS, ECR, ALB), Terraform

Azure & Networking: Azure provisioning with Terraform, Azure Monitor, AWS VPC, load balancing

CI/CD & Containers: Jenkins, GitHub Actions, Docker, Kubernetes, Helm

Monitoring & Operations: Prometheus, Grafana, CloudWatch, Azure Monitor, Linux

Programming & Automation: Python, Bash, Java, TypeScript, SQL

Development & Testing: React, Vite, FastAPI, boto3, REST APIs, pytest, Git

Security & Access: AWS IAM, Kubernetes RBAC, secrets management, GitHub OIDC

EXPERIENCE

DevOps Engineer Jan 2025 – Aug 2026

Neosoft Systems & Cloud Services *Remote*

- Managed and optimized Jenkins and GitHub Actions CI/CD pipelines across development, testing and production environments, reducing manual release effort by approximately 40% through automated build, testing and deployment workflows.
- Provisioned and maintained 3–5 AWS environments using Terraform, managing EC2, S3, IAM, RDS, VPC networking and load balancers to ensure consistent, repeatable infrastructure deployments.
- Supported containerized applications using Docker and Kubernetes, managing deployments, rolling updates, resource limits, health probes and configuration to maintain application stability across production and non-production environments.
- Developed Python and Bash automation scripts for deployment validation, log collection, environment health checks and routine operational tasks, saving approximately 8–10 engineering hours per month.
- Implemented and maintained monitoring dashboards and alerts for 15+ application services using AWS CloudWatch, Prometheus and Grafana, improving infrastructure visibility, incident detection and troubleshooting.
- Investigated and resolved Linux, AWS and Kubernetes infrastructure incidents, analyzing system logs, application metrics, networking issues and container failures to identify root causes and restore service availability.
- Supported scheduled production releases through pre-deployment checks, post-release validation, deployment failure investigation and rollback coordination, maintaining operational runbooks to standardize release and recovery procedures.
- Strengthened cloud and container security through AWS IAM least-privilege policies, Kubernetes RBAC, secrets management and access control practices across application environments.

Software Engineer Nov 2020 – Mar 2023

Zensar Technologies *Pune, India*

- Started in Java application development and transitioned into DevOps responsibilities, gaining hands-on experience in CI/CD automation, cloud infrastructure, containerization and production support.
- Developed and maintained Java applications, troubleshooting defects, optimizing application functionality and supporting testing and release activities.

- Designed and maintained 25+ CI/CD pipelines, automating build, testing and deployment workflows and reducing deployment time from 45 minutes to under 10 minutes.
- Automated AWS and Azure infrastructure provisioning using Terraform, standardizing environment configurations and reducing manual infrastructure setup.
- Supported containerized microservices on Kubernetes using Helm, autoscaling, readiness and liveness probes, and controlled rolling deployments.
- Implemented monitoring and alerting for 20+ application services using Prometheus, Grafana, AWS CloudWatch and Azure Monitor, improving operational visibility and incident detection.
- Investigated application, deployment and infrastructure incidents across Linux, cloud and containerized environments, supporting root cause analysis and service recovery.
- Applied IAM, Kubernetes RBAC, secrets management and vulnerability scanning to strengthen access controls and security across application deployment workflows.

PROJECTS

CloudSpend-Guard – AWS Cost Optimization Aug 2026 – Present
Independent Project – GitHub Repository *AWS / Python / DevOps*

- Engineered an AWS cost optimization platform using Python, FastAPI and boto3 to identify underutilized EC2 instances and unattached EBS volumes through CloudWatch metrics and AWS pricing data.
- Developed a React and TypeScript dashboard to visualize cloud resources, optimization recommendations and estimated cost savings through REST API integration.
- Provisioned AWS infrastructure using Terraform and deployed containerized workloads to private Amazon EKS using Docker, Helm, ECR and an Application Load Balancer.
- Built GitHub Actions CI/CD workflows to automate testing, Docker image builds and ECR publishing, implementing AWS OIDC authentication, least-privilege IAM and immutable Git-SHA image tags.
- Implemented automated backend testing using pytest to validate resource discovery, recommendation logic and API responses.

KubeDiagnose – Kubernetes Diagnostics CLI
Independent Project – GitHub Repository *Kubernetes / Troubleshooting*

- Developed a Python-based Kubernetes diagnostics CLI to investigate workload failures using namespace and workload selection, simplifying troubleshooting of containerized applications.
- Implemented reproducible troubleshooting scenarios covering CrashLoopBackOff, image pull failures, OOMKilled containers, failed health probes and pod scheduling issues.
- Designed diagnostic workflows to investigate container startup failures, resource constraints and Kubernetes workload configuration issues.
- Documented CLI commands, troubleshooting procedures and reproducible failure scenarios with screenshots to demonstrate practical Kubernetes debugging.

CERTIFICATION

AWS Certified DevOps Engineer – Professional

EDUCATION

National College of Ireland Dublin, Ireland
MSc in Cloud Computing 2023 – 2024